

Theorem 9.2.3

If n and r are integers and $1 \leq r \leq n$, then the number of r -permutations of a set of n elements is given by the formula

$$P(n, r) = n(n - 1)(n - 2) \cdots (n - r + 1) \quad \text{first version}$$

or, equivalently,

$$P(n, r) = \frac{n!}{(n - r)!} \quad \text{second version.}$$